

WR-IV

Response-Conditioned Global Completion and the Status of the Past

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Abstract

This paper separates three questions about data and the past: whether a globally compatible realization exists, which historical projections remain compatible, and how a probability law on those histories changes. In the WREH language, a declared world class is restricted by response data and then projected onto hidden histories and observable records. A full-profile realizability defect gives an empty historical class only relative to the declared model. Accumulated hard constraints give nested completion sets without modifying any retained history. Topological response walls transfer to historical families under a homeomorphism over the response base; information-losing projections can erase them. Two classical benchmarks make the distinctions explicit. The doubling map has a unique forward orbit but a Cantor family of full backward histories. For a diffusion with a deterministic initial state, an explicitly constructed regular Brownian-bridge kernel reduces interior variance from $2\kappa t$ to $2\kappa t(T - t)/T$, which remains positive for $0 < t < T$. Exact endpoint data restrict full paths, while the compatible paths on any fixed earlier interval remain unchanged and their probability laws remain equivalent to the prior restrictions. Gaussian noisy observations also preserve measure equivalence. A worked two-observation example verifies the classical Rauch–Tung–Striebel recursion against batch conditioning. The contribution is a controlled framework exposition, with no new theorem-priority claim, physical past rewriting, or identification of evidence order with physical time.

Keywords: global realizability; history projection; non-identifiability; natural extension; regular conditional law; Brownian bridge; fixed-interval smoothing.

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1 Introduction and scope

An observation can improve an estimate of an earlier state without selecting a unique earlier history. It can also exclude complete trajectories while excluding none of their restrictions to a shorter interval. These statements concern different mathematical objects. Treating every reduction of uncertainty as a reduction of a set of possible histories obscures the distinction between compatibility, identifiability and probability.

The question addressed here is conditional and model-relative: given a declared carrier of complete realizations, response data and a historical projection, what can be said about the resulting historical image? A “world” is simply an element of that carrier. It need not be a cosmological universe. The carrier can consist of paths of a stochastic process, deterministic orbits, or other complete objects for which the supplied temporal description is meaningful. Probability is additional structure on a carrier, not a consequence of calling its elements worlds.

1.1 The upstream interface

WR-I supplies response equivalence and the distinction between identifiable quantities and hidden representatives [1, Sections 3–5]. WR-II asks whether compatible finite response data possess a global realization [2, Sections 3–7]. WR-III studies the geometry of response fibres and failure of local triviality [3, Sections 8–11]. WR-IV retains those interfaces and declares a historical projection. It uses the WR-II existence gate before discussing historical ambiguity and the WR-III topology only when a stated family-level bridge is available.

The temporal decomposition is supplied by the model. Neither a set-theoretic response map nor a refinement order generates physical time. The term “past” below always refers to the declared projection and temporal cut. Changing the cut changes the question; changing the carrier changes the model. These changes are not included in the refinement theorem.

1.2 Contribution and claim ceiling

The contribution is an explicit organization of existence, projected compatibility, observable rigidity and probabilistic smoothing within one declared interface. Elementary image properties are proved for completeness. The natural-extension, Gaussian-conditioning, Brownian-bridge and smoothing benchmarks are classical. They are included to test the interface and to identify situations where a plausible verbal inference fails. No priority is claimed for those results or for the general idea of retrospective inference. Section 11 gives a source and non-duplication map.

No theorem here establishes physical creation, disappearance or rewriting of a past, retrocausality, branching ontology, or a physical interpretation of a realizability wall. Mathematical admissibility is not a proof of physical existence. The diffusion is an idealized inverse-problem benchmark, not experimental evidence for WREH as a physical theory. The practical use of the framework is to require an author to name the object that becomes more constrained and the assumptions that justify that statement.

2 Completion sets, historical images and realizability

Let $\mathcal{W}$ be a declared set of complete realizations and let D denote data interpreted as hard constraints. Define the globally compatible class

$$\mathfrak{C}(D) = \{W \in \mathcal{W} : W \text{ is compatible with } D\}. \quad (1)$$

Compatibility includes all constraints of the declared model, not merely agreement with a selected finite list of responses. Assume a total historical projection

$$\pi_- : \mathcal{W} \longrightarrow \mathcal{H}_-.$$

Definition 2.1 (Past-completion set and rigidity). *The past-completion set is $\mathfrak{H}_-(D) = \pi_-(\mathfrak{C}(D))$. Data are world-rigid if $\mathfrak{C}(D)$ is a singleton and past-rigid if $\mathfrak{H}_-(D)$ is a singleton. An empty set is not a rigid set.*

Proposition 2.2 (World rigidity and past rigidity). *World rigidity implies past rigidity. The converse need not hold.*

Proof. The image of a singleton under a total map is a singleton. For the converse, take $\mathcal{W} = \{(h_0, 0), (h_0, 1)\}$, $\mathfrak{C}(D) = \mathcal{W}$ and $\pi_-(h_0, j) = h_0$. The historical image is a singleton although two worlds remain. $\square$

2.1 The WR-II realizability gate

For a protocol family $\mathcal{P}$, let $R : \mathcal{W} \rightarrow \mathcal{Y} = \prod_{P \in \mathcal{P}} Y_P$ be the full response map. For finite $\mathcal{A} \subseteq \mathcal{P}$, denote the restricted map by $R_{\mathcal{A}}$ and the restriction of a profile y by $y_{\mathcal{A}}$. The inherited WR-II objects are

$$\begin{aligned} \mathfrak{C}_{\text{fin}} &= \{y : y_{\mathcal{A}} \in R_{\mathcal{A}}(\mathcal{W}) \text{ for every finite } \mathcal{A} \subseteq \mathcal{P}\}, \\ \mathfrak{D}_{\text{fin}} &= \mathfrak{C}_{\text{fin}} \setminus R(\mathcal{W}). \end{aligned} \quad (2)$$

Full-profile data D_y mean $\mathfrak{C}(D_y) = R^{-1}(y)$. Finite-profile data are a different object.

Proposition 2.3 (Existence precedes historical ambiguity). *For full-profile data and a total π_- ,*

$$\mathfrak{H}_-(D_y) = \emptyset \iff R^{-1}(y) = \emptyset \iff y \notin R(\mathcal{W}). \quad (3)$$

In particular, $y \in \mathfrak{D}_{\text{fin}}$ has no compatible historical image in the declared model.

Proof. An image under a total map is empty exactly when its domain is empty. A fibre is empty exactly when its base point is outside the image. The last assertion uses (2). No additional existence theorem is supplied by historical projection. $\square$

Example 2.4 (Every finite record can fit while the full profile fails). Use the WR-II finite-support model [2, Section 7]: $\mathcal{W}$ consists of binary sequences with finitely many ones, and $R_n(W) = W_n$. The target $y = (1, 1, \dots)$ is realized on every finite coordinate set by putting ones on those coordinates and zeros elsewhere. No finite-support sequence realizes the full target. Thus the full historical image is empty for every total historical projection, although every finite-data completion set is nonempty. This example demonstrates why the full-profile qualifier in Proposition 2.3 is necessary.

Remark 2.5 (Empty, ambiguous and probability-zero are distinct). An empty completion class diagnoses incompatibility relative to $\mathcal{W}$. A nonempty historical image with more than one element diagnoses historical ambiguity relative to π_- . A nonempty fibre can have prior probability zero; exact Brownian endpoints provide an example. “Ghost past” can serve as a mnemonic for model-relative emptiness, but it is not a physical conclusion and will not replace the mathematical condition.

3 Refinement, observable records and resolution

Throughout refinement, fix $\mathcal{W}$, π_- and the temporal cut. Write $D \preceq D'$ if D' retains all hard constraints of D and possibly adds further ones. Revising a measurement, enlarging the model class, or moving the cut is not refinement in this sense.

Theorem 3.1 (Completion refinement).

$$D \preceq D' \implies \mathfrak{C}(D') \subseteq \mathfrak{C}(D), \quad \mathfrak{H}_-(D') \subseteq \mathfrak{H}_-(D).$$

Proof. A world satisfying every constraint of D' satisfies every constraint of D . Images under the same map preserve inclusion. Both inclusions may be equalities; the smaller sets may be empty. $\square$

Corollary 3.2 (Consistent refinement preserves rigidity). *If $\mathfrak{H}_-(D) = \{h\}$, $D \preceq D'$ and $\mathfrak{C}(D') \neq \emptyset$, then $\mathfrak{H}_-(D') = \{h\}$.*

Proof. The refined historical image is a nonempty subset of a singleton. $\square$

Remark 3.3 (Non-rewriting principle). For each retained $W \in \mathfrak{C}(D')$, the value $\pi_-(W)$ is the same before and after refinement. Set restriction removes incompatible objects; it does not alter any retained object. This is a statement about the mathematical operation, not a theorem about all possible physical mechanisms.

3.1 Hidden histories and observable histories

Let a fixed map $Q_- : \mathcal{H}_- \rightarrow \mathcal{O}_-$ encode accessible historical records and put $\mathfrak{D}_-(D) = Q_-(\mathfrak{H}_-(D))$. Such records need not retain all hidden information.

Proposition 3.4 (Observable rigidity). *$\mathfrak{D}_-(D)$ is a singleton exactly when $\mathfrak{H}_-(D)$ is nonempty and Q_- is constant on it. Observable rigidity can coexist with hidden-past ambiguity.*

Proof. A nonempty set has a singleton image exactly when the map is constant on that set. For ambiguity, take two distinct histories h_1, h_2 with $Q_-(h_1) = Q_-(h_2)$ and $\mathfrak{H}_-(D) = \{h_1, h_2\}$. Agreement for one pair would not suffice if additional histories had different records. $\square$

For example, two particle paths can agree at every declared record time and differ between those times. Their record histories agree even when their continuous hidden histories do not. This is a modelling choice about access, not a proof that the unrecorded segments have no physical value.

3.2 Finite resolution does not automatically define an equivalence

For a metric record space $(\mathcal{O}_-, d)$ and tolerance $\varepsilon > 0$, consider the relation

$$d(Q_-(h), Q_-(h')) \leq \varepsilon.$$

It need not be transitive. Records $0, 0.75\varepsilon, 1.5\varepsilon$ give an immediate counterexample. This is the WR-I tolerance warning [1, Section 8]; exact observational classes and finite-resolution ambiguity are also explicitly distinguished in recent measurement-design work [11, Sections III and V].

A declared quantizer $q : \mathcal{O}_- \rightarrow \mathcal{B}$ does give an equivalence by equality of bin labels. This equivalence depends on the binning and should not be confused with the pairwise metric-tolerance relation. Likewise, a statement that a historical diameter decreases requires a declared metric, and a statement that its probability uncertainty decreases requires a declared measure and functional. Cardinality alone supplies neither.

4 Topological historical families

Suppose $\mathcal{W}$, $\mathcal{H}_-$ and $E = R(\mathcal{W})$ carry declared topologies and that R and π_- are continuous. For open $U \subseteq E$, form the response-indexed historical image family

$$\mathcal{H}_U = \{(y, h) \in U \times \mathcal{H}_- : h \in \pi_-^{-1}(R^{-1}(y))\}, \quad p(y, h) = y. \quad (4)$$

Use the subspace topology from $U \times \mathcal{H}_-$. Local triviality is not assumed. A point is structurally regular when its family admits a local fibre-bundle trivialization; the complementary locus is the WR-III structural wall [3, Section 8]. The word wall supplies no codimension, smoothness or physical transition by itself.

Proposition 4.1 (Conditional transfer of local triviality). *Consider the map*

$$F : R^{-1}(U) \rightarrow \mathcal{H}_U, \quad F(W) = (R(W), \pi_-(W)).$$

If F is a homeomorphism, then $R : R^{-1}(U) \rightarrow U$ and $p : \mathcal{H}_U \rightarrow U$ have the same local-triviality locus.

Proof. Since $p \circ F = R$, a trivialization $\phi : R^{-1}(V) \rightarrow V \times A$ for open $V \subseteq U$ transfers to $\phi \circ F^{-1} : p^{-1}(V) \rightarrow V \times A$. It still commutes with the base projections. Conversely use F . Thus a trivialization exists for one family exactly when it exists for the other. $\square$

A collection of separate homeomorphisms between individual fibres does not establish a homeomorphism of total families. The stated hypothesis is sufficient, not claimed necessary. Also, (4) is not automatically a pullback bundle. A genuine pullback starts from a bundle $q : Z \rightarrow B$ and a map $f : U \rightarrow B$, giving $f^*Z = \{(u, z) : f(u) = q(z)\}$ [10, Section 1.2, p. 7]. Those data are absent from the image-family definition.

Example 4.2 (A projection can erase a response wall). The WR-III sphere-height response $R : S^2 \rightarrow [-1, 1]$, $R(x, y, z) = z$, has circular fibres for $-1 < z < 1$ and singleton fibres at $z = \pm 1$ [3, Section 10]. The endpoints are wall points: every relative neighbourhood includes fibres of different topological types. If π_- is constant with value h_0 , the historical family is $[-1, 1] \times \{h_0\}$ and is trivial everywhere. The world wall disappears under historical projection.

This is an information-loss example without an assigned physical chronology. A response map is not thereby a forward-evolution map. The circle and the singleton are both connected, so even a visible wall need not change the number of connected components. Refinement of hard constraints and transfer of local-triviality walls are different operations.

5 Deterministic dynamics and full backward histories

Let $T : X \rightarrow X$ be a deterministic map. A full backward history ending at x_0 is a sequence $(x_0, x_{-1}, x_{-2}, \dots)$ with $T(x_{-n}) = x_{-(n-1)}$ for all $n \geq 1$. Define

$$\hat{X} = \{(x_0, x_{-1}, \dots) \in X^{\mathbb{N}_0} : T(x_{-n}) = x_{-(n-1)}, n \geq 1\}, \quad p_0(\hat{x}) = x_0. \quad (5)$$

For a continuous map, use the subspace topology inherited from the product. Histories over one present state form $p_0^{-1}(x_0)$, not the entire space $\hat{X}$. The natural-extension shift sends $(x_0, x_{-1}, \dots)$ to $(T(x_0), x_0, x_{-1}, \dots)$; its inverse deletes the first coordinate. Thus the history-space dynamics is invertible even when T on present states is not. This construction is classical; the circle-doubling case is the dyadic solenoid [6, Section 1].

Proposition 5.1 (Forward uniqueness is not backward uniqueness). *The forward orbit of each x_0 is unique. Noninjectivity of T does not guarantee existence or multiplicity of full backward histories over every x_0 .*

Proof. The future is $(x_0, T(x_0), T^2(x_0), \dots)$. For a counterexample to automatic backward multiplicity, let $X = \mathbb{N}_0$, $T(0) = 1$ and $T(n) = n$ for $n \geq 1$. No history ends at 0. Over 1, the only full history is $(1, 1, \dots)$ because choosing predecessor 0 prevents further extension. The next section gives genuine multiple full histories. $\square$

6 The doubling-map benchmark

Take $X = \mathbb{R}/\mathbb{Z}$ with its circle topology and $T(x) = 2x \pmod{1}$. Let the world carrier be $\hat{X}$, the present response be p_0 , and the historical projection retain all negative coordinates. Each state has two immediate predecessors, but the following assertion concerns complete infinite histories.

Theorem 6.1 (Cantor family over an exactly known present). *For every $x_0 \in X$, the fibre $p_0^{-1}(x_0)$ is homeomorphic to $\{0, 1\}^{\mathbb{N}}$ with its product topology.*

Proof. Fix the representative $r_0 \in [0, 1)$ of x_0 . For a bit sequence ϵ , define

$$r_n = \frac{r_{n-1} + \epsilon_n}{2}, \quad x_{-n} = [r_n].$$

These coordinates form a full history. Conversely, the unique representatives $r_n \in [0, 1)$ of a history recover $\epsilon_n = 2r_n - r_{n-1} \in \{0, 1\}$. The correspondence is bijective. Each coordinate depends on finitely many bits, so the coding is continuous. The domain is compact and the target fibre is Hausdorff; a continuous bijection is therefore a homeomorphism. The branch labels are chosen over this fixed x_0 ; no globally continuous inverse branch on the circle is asserted. $\square$

Corollary 6.2 (Finite branch records leave infinitely many choices). *Fixing any finite number of branch bits leaves a Cantor family of full histories.*

Proof. Removing the fixed bit coordinates leaves countably infinitely many free coordinates, whose product is again $\{0, 1\}^{\mathbb{N}}$. For a fixed initial block these encode progressively older predecessors. $\square$

At depth n there are 2^n distinct finite predecessor chains. A picture at finite depth is not a representation of all infinite histories. The Cantor assertion is topological and set-theoretic; it does not assign probabilities to branch choices and does not establish a physical branching ontology.

7 The probability layer and Gaussian conditioning

A probability law on $\mathcal{W}$ adds relative weights to realizations. It must be supplied separately from compatibility. Conditional laws on continuous observations require a kernel, because a singleton observed value can have probability zero.

Definition 7.1 (Regular conditional kernel). *For measurable random elements W, Y with laws μ, ν , a kernel K_y is a regular conditional law of W given Y if K_y is a probability measure for each y , $y \mapsto K_y(F)$ is measurable for each measurable F , and*

$$\mathbb{P}(W \in F, Y \in A) = \int_A K_y(F) \nu(dy). \quad (6)$$

The identity determines a kernel only ν -almost everywhere. Statements at a specified zero-probability value need a declared version or a regularity condition selecting one. We construct such a version for the Brownian endpoint, rather than divide by a zero probability.

7.1 Gaussian retrospective contraction

Fix earlier data $A = a$ and work throughout under their declared conditional law. Suppose a past state $X \in \mathbb{R}^n$ and additional data $Z \in \mathbb{R}^m$ are jointly Gaussian under that law, with means m_X, m_Z and covariance

$$\begin{pmatrix} P & C \\ C^\top & S \end{pmatrix}, \quad S \succ 0.$$

Theorem 7.2 (Classical Gaussian covariance reduction). *A continuous conditional Gaussian kernel is*

$$X \mid Z = z \sim \mathcal{N}(m_X + CS^{-1}(z - m_Z), P_s), \quad P_s = P - CS^{-1}C^\top. \quad (7)$$

Degenerate Gaussian laws are allowed. Moreover,

$$0 \preceq P_s \preceq P, \quad \Delta P = P - P_s = CS^{-1}C^\top \succeq 0. \quad (8)$$

For any $v \in \mathbb{R}^n$, $v^\top \Delta P v > 0$ exactly when $C^\top v \neq 0$.

Proof. Set $E = X - m_X - CS^{-1}(Z - m_Z)$. Direct multiplication gives $\text{Cov}(E, Z) = 0$. Joint Gaussianity, including possible degeneracy, implies independence of E and Z . Its covariance is $P - CS^{-1}C^\top$, hence positive semidefinite. Translation of the independent residual gives (7) and a measurable, weakly continuous version. Since $S^{-1} \succ 0$,

$$v^\top \Delta P v = (C^\top v)^\top S^{-1}(C^\top v)$$

is positive precisely in the stated case. This is the standard Gaussian conditioning formula, also given in [9, Appendix A.1, Lemma A.3]. $\square$

The assumption $S \succ 0$ is used to invert the observation covariance. No extension by an unspecified inverse at singular S is intended. A singular-observation treatment must restrict to the attainable observation subspace and specify its kernel separately. A zero cross-covariance $C = 0$ gives no reduction. Positive residual covariance is not guaranteed by the general theorem; for example, observing $Z = X$ exactly can determine X .

7.2 What variance monotonicity does and does not say

For nondegenerate scalar Gaussians, central intervals of a fixed probability become shorter when variance decreases. Their centres can move, so the intervals need not be nested. Scalar supports remain $\mathbb{R}$ as long as both variances are positive. These facts already rule out identifying covariance reduction with set restriction.

Proposition 7.3 (General conditional variance decreases only on average). *Let X be scalar and square-integrable, and let $\mathcal{G} \subseteq \mathcal{F}$ be sigma-fields. Then*

$$\text{Var}(X \mid \mathcal{G}) = \mathbb{E}[\text{Var}(X \mid \mathcal{F}) \mid \mathcal{G}] + \text{Var}(\mathbb{E}[X \mid \mathcal{F}] \mid \mathcal{G}) \quad a.s. \quad (9)$$

Consequently the first term is at most the left side, but $\text{Var}(X \mid \mathcal{F})$ itself need not be pointwise smaller.

Proof. Write $X - \mathbb{E}[X \mid \mathcal{G}] = (X - \mathbb{E}[X \mid \mathcal{F}]) + (\mathbb{E}[X \mid \mathcal{F}] - \mathbb{E}[X \mid \mathcal{G}])$. The conditional expectation of the cross term given $\mathcal{G}$ is zero by the tower property. Squaring and taking conditional expectations gives (9). $\square$

Example 7.4 (A rare observation increases conditional variance). Let $\mathbb{P}(X = 0) = 0.99$ and $\mathbb{P}(X = 10) = \mathbb{P}(X = -10) = 0.005$. The prior mean is zero and variance is 1. Observing $Z = \mathbf{1}_{\{X \neq 0\}}$ gives variance 100 when $Z = 1$ and zero when $Z = 0$. The average posterior variance remains 1. Thus the Gaussian pointwise conclusion must not be exported to arbitrary observations.

8 Exact-endpoint diffusion and regular Brownian bridges

Fix $x_0 \in \mathbb{R}$, $T > 0$ and diffusivity $\kappa > 0$. Let B be standard Brownian motion and

$$X_t = x_0 + \sqrt{2\kappa} B_t, \quad 0 \leq t \leq T. \quad (10)$$

The initial state is deterministic. With only this state fixed, $X_t \sim \mathcal{N}(x_0, 2\kappa t)$; this is the prior relative to the endpoint smoothing problem. The coefficient κ has units of length squared per time; D continues to denote evidence. The model is an idealized diffusion, not a fitted experimental system.

8.1 The WREH dictionary

Use the Polish carrier $\mathcal{W}_T = C_{x_0}([0, T], \mathbb{R})$ of continuous paths starting at x_0 , with uniform topology and Borel sigma-field. Its prior path law is μ . Fix $0 < \tau < T$.

Object	Declaration
World and carrier	$w \in \mathcal{W}_T$; μ supplies probability weights.
Protocol / response	$R_{P_T}(w) = w(T)$.
Data	D_b : the exact endpoint b ; x_0 is part of the carrier.
Completion fibre	$\mathfrak{C}(D_b) = \{w \in \mathcal{W}_T : w(T) = b\} =: C_b$.
Full pre-endpoint past	$\pi_-(w) = w _{[0, T]}$, codomain $\pi_-(\mathcal{W}_T)$.
Earlier fragment	$\pi_\tau(w) = w _{[0, \tau]}$, codomain $C_{x_0}([0, \tau], \mathbb{R})$.
Conditional law	K_b , a measure with $K_b(C_b) = 1$, not the set C_b itself.

The closed path space is Polish because it is a closed subspace of the continuous-function Banach space. The definition of admissible paths is distinct from the probability of an individual path. In particular $C_b \neq \emptyset$ although $\mu(C_b) = 0$.

8.2 An explicit version at every endpoint

Proposition 8.1 (Regular Brownian-bridge kernel). *For every $b \in \mathbb{R}$, let K_b be the law on $\mathcal{W}_T$ of*

$$U_t^b = x_0 + \frac{t}{T}(b - x_0) + \sqrt{2\kappa} \left(B_t - \frac{t}{T}B_T \right), \quad 0 \leq t \leq T. \quad (11)$$

Then $b \mapsto K_b$ is weakly continuous, is a regular conditional law of X given X_T , and satisfies $K_b(C_b) = 1$ for every b . Its covariance is

$$\text{Cov}_{K_b}(X_s, X_t) = 2\kappa \left(\min(s, t) - \frac{st}{T} \right). \quad (12)$$

Proof. Put $V_t = B_t - tB_T/T$. For every t , $\text{Cov}(V_t, B_T) = t - t = 0$. Each finite-dimensional vector of V is jointly Gaussian with B_T and independent of it. Evaluations at a countable dense set of times generate the Borel sigma-field of continuous paths; hence the entire residual path is independent of B_T . Its covariance is $\min(s, t) - st/T$. Also,

$$X_t = x_0 + \frac{t}{T}(X_T - x_0) + \sqrt{2\kappa} V_t.$$

Independence gives (6) with $W = X$, $Y = X_T$ and $\nu = \mathcal{N}(x_0, 2\kappa T)$. The map $(b, v) \mapsto x_0 + t(b - x_0)/T + \sqrt{2\kappa} v_t$ is continuous into path space. Its pushforward kernel is measurable; dominated convergence for bounded continuous path functionals proves weak continuity in b . At 0 and T , the residual vanishes, so the path starts at x_0 and ends at b . Covariance gives (12). The residual construction is classical [7, Theorem 22.1 and following remark]. $\square$

The notation $X \mid X_T = b$ will always mean this selected kernel. An arbitrary regular conditional version can be changed at a single b without affecting disintegration. Weak continuity, together with the full support of the endpoint Gaussian law, removes that ambiguity among weakly continuous versions. There is no division by $\mathbb{P}(X_T = b)$.

8.3 Variance reduction with residual path ambiguity

Theorem 8.2 (The interior state is sharpened but not determined). *For the kernel above and every $0 < t < T$,*

$$X_t \mid X_T = b \sim \mathcal{N} \left(x_0 + \frac{t}{T}(b - x_0), 2\kappa \frac{t(T-t)}{T} \right), \quad (13)$$

$$0 < 2\kappa \frac{t(T-t)}{T} < 2\kappa t.$$

Proof. Equation (11) gives the mean and Gaussianity, while (12) gives the variance. Independently, $\text{Cov}(X_t, X_T) = 2\kappa t$ and $\text{Var}(X_T) = 2\kappa T$, so Theorem 7.2 gives

$$2\kappa t - \frac{(2\kappa t)^2}{2\kappa T} = 2\kappa \frac{t(T-t)}{T}.$$

Since $0 < (T-t)/T < 1$, both inequalities follow. At $t = 0$ and $t = T$ the bridge variance is zero; these boundaries are excluded from the strict assertion. $\square$

Corollary 8.3 (A countable list of paths has zero bridge probability). *K_b is not a Dirac law and assigns zero probability to any countable family of trajectories. In particular the compatible full path class remains uncountable.*

Proof. Fix $t_* \in (0, T)$. Every specified path imposes one value at t_* , an event of probability zero under its nondegenerate Gaussian marginal. Countable unions remain null. Since $K_b(C_b) = 1$, C_b cannot be countable. $\square$

8.4 Full histories and fixed earlier fragments

Proposition 8.4 (A smaller full fibre can have the same earlier image). *C_b is a proper subset of $\mathcal{W}_T$. The restriction π_- to $[0, T]$ is injective, and $\pi_-(C_b)$ is a proper subset of $\pi_-(\mathcal{W}_T)$ characterized by limit b at T . For every fixed $0 < \tau < T$, however,*

$$\pi_\tau(C_b) = \pi_\tau(\mathcal{W}_T) = C_{x_0}([0, \tau], \mathbb{R}). \quad (14)$$

Proof. A linear path to b lies in C_b , whereas a linear path to any $b' \neq b$ does not. Continuous paths agreeing before T agree at T by taking limits, proving injectivity. For a given prefix h , keep it on $[0, \tau]$ and put $w(t) = h(\tau) + (t - \tau)(b - h(\tau))/(T - \tau)$ for $\tau \leq t \leq T$. This continuous extension proves (14). $\square$

One can say more about the probability laws of these fragments without confusing them with the sets. Put $\mu_\tau = (\pi_\tau)_*\mu$, $K_b^\tau = (\pi_\tau)_*K_b$, and define the transition density

$$p_s(z) = \frac{1}{\sqrt{4\pi\kappa s}} \exp\left(-\frac{z^2}{4\kappa s}\right), \quad s > 0.$$

Proposition 8.5 (Equivalent laws on each fixed earlier interval). *For every $b \in \mathbb{R}$ and $0 < \tau < T$,*

$$\frac{dK_b^\tau}{d\mu_\tau}(h) = \frac{p_{T-\tau}(b - h(\tau))}{p_T(b - x_0)} > 0. \quad (15)$$

The two measures are equivalent, with the same null sets and topological support. Nevertheless their state variances differ as in (13).

Proof. Independent increments imply that, given the prefix h , the endpoint has density $p_{T-\tau}(b - h(\tau))$. Its marginal density is $p_T(b - x_0)$. Their ratio defines a normalized prefix kernel by disintegration. For bounded continuous prefix functionals its integral is continuous in b : on compact b -sets the denominator has a positive lower bound and the numerator is bounded. This version agrees almost everywhere with the restriction of the kernel in Proposition 8.1; weak continuity and full support of the endpoint law make them agree everywhere. The strictly positive density gives mutual absolute continuity. Equivalent measures give positive mass to the same open sets, hence have the same support. $\square$

On the full interval $[0, T]$ the laws K_b and μ are instead singular, since $K_b(C_b) = 1$ and $\mu(C_b) = 0$. Thus extending the cut all the way to T changes the measure-theoretic conclusion. Equivalence on every fixed shorter interval is consistent with singularity on the full path space. Positive variance alone would not prove any of these support or equivalence statements.

9 Noisy and finite-resolution endpoint data

Let $Y_T = X_T + \eta$ with independent $\eta \sim \mathcal{N}(0, R)$, $R > 0$. Here R without a subscript denotes scalar noise variance, while R_{P_T} is the response map. The units of R are length squared.

Proposition 9.1 (Classical noisy smoothing law). *For every observed y in the continuous Gaussian version and $0 < t < T$,*

$$\mathbb{E}[X_t \mid Y_T = y] = x_0 + \frac{2\kappa t}{2\kappa T + R}(y - x_0), \quad (16)$$

$$\text{Var}(X_t \mid Y_T) = 2\kappa t \left(1 - \frac{2\kappa t}{2\kappa T + R}\right). \quad (17)$$

This variance lies strictly between the exact-bridge variance and the prior variance. It approaches them as $R \downarrow 0$ and $R \rightarrow \infty$, respectively.

Proof. The jointly Gaussian pair has $\text{Cov}(X_t, Y_T) = 2\kappa t$ and $\text{Var}(Y_T) = 2\kappa T + R$. Apply (7). Since $2\kappa T + R > 2\kappa T > 2\kappa t > 0$, the inequalities and limits follow. $\square$

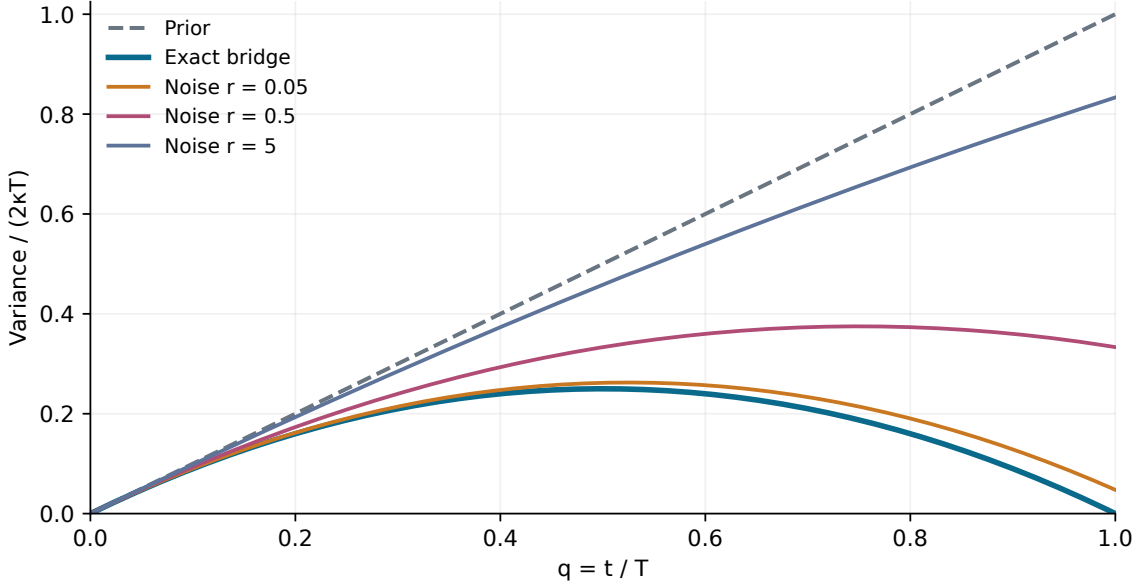


Figure 1: Analytic normalized state variance. With $q = t/T$ and $r = R/(2\kappa T)$, the prior gives q , an exact endpoint gives $q(1 - q)$, and a Gaussian noisy endpoint gives $q - q^2/(1 + r)$. Interior exact-bridge variance is positive. Curves are formulas, not empirical estimates.

9.1 Likelihood changes weights without necessarily removing paths

For each y the likelihood and normalizer are

$$L_y(w) = \frac{1}{\sqrt{2\pi R}} e^{-(y-w(T))^2/(2R)}, \quad Z_y = \int L_y(w) \mu(dw).$$

Proposition 9.2 (Equivalent noisy posterior). $0 < Z_y < \infty$, and $d\mu_y/d\mu = L_y/Z_y$ defines an everywhere specified regular conditional path law given $Y_T = y$. It is equivalent to μ and has the same topological support.

Proof. The likelihood is strictly positive and bounded; μ is a probability measure. Conditioning the independent observation noise on each path and integrating proves disintegration with endpoint-observation density Z_y . Strict positivity gives $\mu_y(A) = 0$ exactly when $\mu(A) = 0$. The support statement follows as in Proposition 8.5. This is the standard likelihood-based Bayesian inverse-problem construction [5, Section 2 and Theorem 6.31], specialized here and proved directly. $\square$

There is a corresponding hard-compatibility calculation, provided the noise variable is included in the worlds. For a declared admissible noise set $S \subseteq \mathbb{R}$,

$$\begin{aligned} J_y^S &= \{(w, \eta) \in \mathcal{W}_T \times S : w(T) + \eta = y\}, \\ \text{pr}_w(J_y^S) &= \{w \in \mathcal{W}_T : y - w(T) \in S\}. \end{aligned} \tag{18}$$

Gaussian noise has full support, and the declared set $S = \mathbb{R}$ permits every residual. The projected compatibility class is all of $\mathcal{W}_T$. If instead $S = [-a, a]$ with $a > 0$, the image is $\{w : |w(T) - y| \leq a\}$, an endpoint band. It is not a tube constraining the whole path. The bound a is an amplitude, distinct from variance R . Support-based admissibility does not require positive probability for individual noise values.

9.2 A finite-resolution hard observation

A different model records that the true endpoint lies in $I = [b - a, b + a]$, $a > 0$, without introducing additive noise. The compatible set is $C_I = \{w : w(T) \in I\}$. The event has positive probability and the

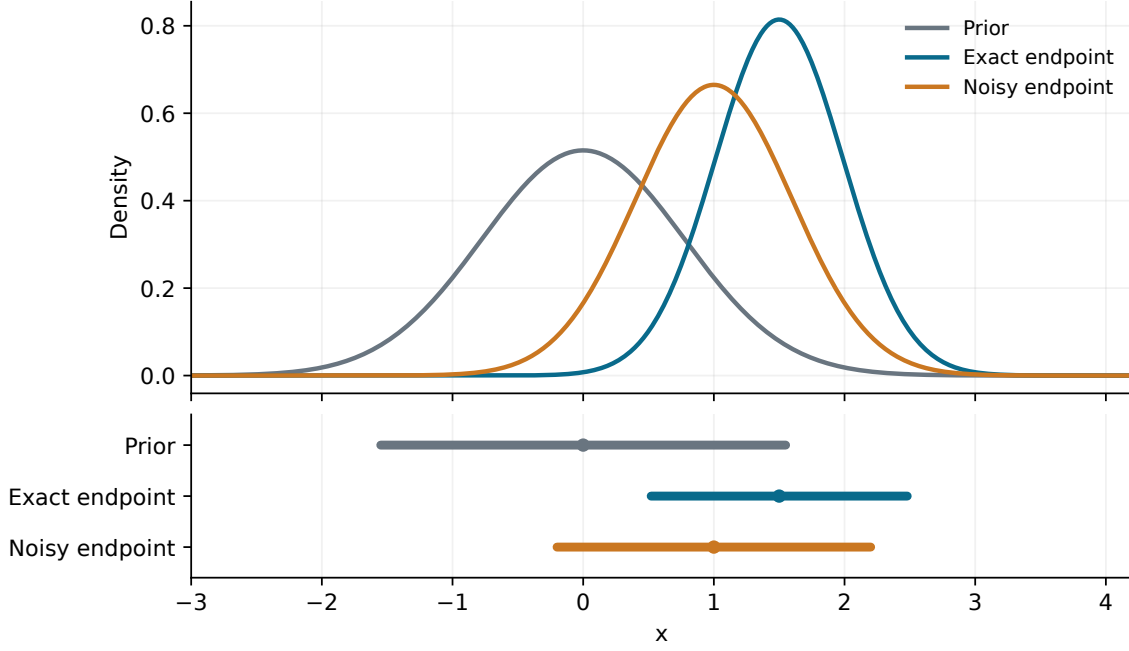


Figure 2: A later observation both shifts and narrows the earlier marginal. Parameters: $x_0 = 0$, $\kappa = 1/2$, $T = 1$, $t = 0.6$, endpoint datum 2.5, and noisy variance $R = 0.5$. Means and variances are $(0, 0.6)$, $(1.5, 0.24)$ and $(1, 0.36)$. Segments mark central intervals $m \pm 2\sigma$ (about 95.45% probability), whose lengths decrease without nesting. Every displayed density has support $\mathbb{R}$.

ordinary conditional law is

$$\mu(\cdot \mid X_T \in I) = \frac{\int_I K_z(\cdot) p_T(z - x_0) dz}{\int_I p_T(z - x_0) dz}. \quad (19)$$

Every fixed earlier prefix still extends to an endpoint in I . Formula (19) is a mixture of bridges, not a bridge pinned to the midpoint. Its variance must not be replaced by the exact-endpoint formula. Shrinking a to zero gives the selected kernel weakly, by its weak continuity. Bounded measurement noise and a true-endpoint interval have related set projections but different likelihood models.

10 Several observation times: a classical RTS example

The endpoint benchmark extends to several observations by ordinary Gaussian smoothing. To state exactly what is being used, sample the diffusion at $0 = s_0 < s_1 < \dots < s_m$, and write

$$X_k = X_{k-1} + \xi_k, \quad Y_k = X_k + \eta_k, \quad q_k = 2\kappa(s_k - s_{k-1}) > 0, \quad (20)$$

where $X_0 = x_0$ is deterministic, $\xi_k \sim \mathcal{N}(0, q_k)$, $\eta_k \sim \mathcal{N}(0, R_k)$, $R_k > 0$, and all noises are mutually independent. The scalar random-walk specialization is sufficient here; no nonlinear smoothing method is proposed.

10.1 Forward filtering and backward smoothing

Let m_k, p_k be the mean and variance of X_k given $y_1, \dots, y_k$, starting from $m_0 = x_0, p_0 = 0$. The forward Kalman update in this model is

$$\begin{aligned} m_k^- &= m_{k-1}, & p_k^- &= p_{k-1} + q_k, & K_k &= \frac{p_k^-}{p_k^- + R_k}, \\ m_k &= m_k^- + K_k(y_k - m_k^-), & p_k &= (1 - K_k)p_k^-. \end{aligned} \quad (21)$$

Initialize $m_m^s = m_m, p_m^s = p_m$. The classical Rauch–Tung–Striebel (RTS) backward recursion [8, Section 3.2][9, Section 12.2, Theorem 12.2] becomes

$$\begin{aligned} G_k &= \frac{p_k}{p_k + q_{k+1}}, \\ m_k^s &= m_k + G_k(m_{k+1}^s - m_k), \\ p_k^s &= p_k + G_k^2(p_{k+1}^s - p_k - q_{k+1}), \quad k = m-1, \dots, 0. \end{aligned} \tag{22}$$

To verify this specialization, conditional on observations through k the pair (X_k, X_{k+1}) has covariance $\begin{pmatrix} p_k & p_k \\ p_k & p_k + q_{k+1} \end{pmatrix}$. Gaussian conditioning gives mean $m_k + G_k(X_{k+1} - m_k)$ and variance $p_k - G_k^2(p_k + q_{k+1})$. Future observations are conditionally independent of X_k given X_{k+1} and the observations through k , by the declared state-space model. Conditional expectation and total variance therefore give (22). All denominators are positive. This is a direct proof of the stated classical specialization, not a new recursion.

10.2 A two-observation calculation and independent batch check

Take $m = 2, x_0 = 0, q_1 = q_2 = R_1 = R_2 = 1$, and $(y_1, y_2) = (0.2, 1.4)$. Filtering and smoothing give

$$m_1 = 0.1, \quad p_1 = 0.5, \quad m_2 = 0.88, \quad p_2 = 0.6,$$

$$G_1 = \frac{1}{3}, \quad m_1^s = 0.36, \quad p_1^s = 0.4.$$

Independently, for X_1 and $Z = (Y_1, Y_2)^\top$ the prior covariance blocks are

$$P = 1, \quad C = (1, 1), \quad S = \begin{pmatrix} 2 & 1 \\ 1 & 3 \end{pmatrix}, \quad S^{-1} = \frac{1}{5} \begin{pmatrix} 3 & -1 \\ -1 & 2 \end{pmatrix}.$$

Thus $CS^{-1} = (2/5, 1/5)$ gives mean $(2/5)0.2 + (1/5)1.4 = 0.36$ and variance $1 - CS^{-1}C^\top = 0.4$, agreeing with RTS. The later observation lowers the earlier variance from 0.5 to 0.4 and changes its mean from 0.1 to 0.36. These are exact finite-dimensional calculations, not fitted simulation results.

With Gaussian measurement noise, the likelihood of any finite observation vector is strictly positive for every path. The posterior path measure is therefore equivalent to the prior, as in Section 9. Including unrestricted noise variables in the carrier likewise leaves every path compatible with the observation vector. If the state observations are instead exact, paths must interpolate the supplied values at the sample times, while the continuous segments between those times remain nonunique. Covariance, path compatibility and sampling resolution must still be reported separately.

11 Prior work, non-duplication and limits

The distinctions used here belong to established mathematical traditions. Kalman’s realization and observability theory relates hidden states to input-output information [4, Sections 4–5]; a record projection that forgets hidden information is not a new observability concept. Bayesian inverse problems explicitly distinguish candidate objects from their posterior probabilities [5, Sections 1–2]. Inverse limits and solenoids supply the deterministic history benchmark. Gaussian smoothing supplies the stochastic one. The present paper brings these into the WREH global-completion interface, rather than replacing their theories.

Ingredient	Inherited / classical	Role in this paper
Response equivalence	WR-I; identifiability; observational quotients.	Restrict the declared carrier, then project to histories.
Full-profile existence	WR-II; finite/global consistency.	Gate historical ambiguity by nonempty completion.
Structural walls	WR-III; local bundle triviality.	State a sufficient transfer condition and show erasure.
Infinite pasts	Natural extensions and dyadic solenoids.	Separate deterministic future from backward ambiguity.
Conditional laws	Gaussian residuals; Brownian bridges; Bayes.	Separate exact fibres, null events, supports and variance.
Several-time smoothing	Classical RTS recursion.	Give a fully specified reproducible application.

Accordingly, the set and image propositions are elementary consequences of definitions, and the wall-transfer result is conjugation of trivializations. The benchmark proofs are self-contained applications of standard mathematics. The controlled contribution is the declared interface and the explicit comparison of several meanings of “the past becomes more definite”. No result in the table is presented as a newly discovered mathematical theorem. A targeted source audit supports the stated results; it cannot certify that the wider architecture is unprecedented in every field.

11.1 A realizability gap is not automatically information gain

WR-II defines $\Gamma(y) = \delta_{\text{glob}}(y) - \delta_{\text{fin}}(y)$ when both normalized residual radii are finite [2, Section 10]. For a finite protocol family the two radii coincide, so $\Gamma = 0$. The two-observation example above nevertheless has positive covariance gain. Moreover, WR-II’s radii depend on declared protocol metrics and scales, while probabilistic information quantities depend on a probability model and reference law. Therefore Γ alone is not a universal surprisal or information gain.

Any comparison must specify which random objects, data law, information functional and calibration are held fixed. Such a comparison remains future work. No information-theoretic identification or next programme branch is established by this preprint.

11.2 Inference order and physical time

Data can be acquired later, processed out of order, reanalysed, or combined with records concerning other times. The order $D \preceq D'$ describes retained constraints. The index t in (10) describes the model’s internal time. No map equating these orders is supplied. This is a scope statement, not an impossibility theorem about all models. Retrodiction is an inference about an earlier index, not a backwards physical influence asserted by the conditioning operation.

12 Conclusions and research questions

A present response, existence of a global completion, uniqueness of a hidden historical projection and uniqueness of accessible records are distinct statements. An empty full-profile fibre signifies model-relative incompatibility. A nonempty ambiguous historical fibre signifies non-identifiability at a specified projection. Neither is interchangeable with a probability-zero event.

Hard-constraint accumulation nests compatible sets without changing retained histories. A local-triviality wall transfers under the stated bridge between families. A projection that loses information can erase a wall. The doubling map provides a classical example of a unique deterministic future with a Cantor family of full pasts. None of these facts chooses a physical ontology.

The diffusion benchmark answers the central question precisely. An exact endpoint reduces interior variance but does not determine a trajectory. It restricts the full path fibre and the full pre-endpoint image.

On every fixed earlier interval, both the compatible path set and the measure-equivalence class remain unchanged. Noisy Gaussian endpoint data preserve full path-measure equivalence. These simultaneous conclusions are compatible because they refer to different objects and cuts.

Open research questions concern model-dependent historical metrics, finite-resolution record structures, projections that retain enough topology for nontrivial wall transfer, and non-Gaussian smoothing with explicit failure criteria. Physical applications require a justified carrier, observation process and temporal model. A stronger novelty claim would require a correspondingly broader literature comparison and a nonduplicative technical result. These are future directions, not missing hypotheses concealed in the present theorems.

A Reproducibility and the supplementary demonstration

The deposit package contains both language versions, their shared LaTeX source, two analytic figures per language, a script regenerating the figures and independently checking RTS against batch conditioning, and a self-contained bilingual HTML demonstration. The HTML runs locally without network requests. Its plots show analytic marginal laws and finitely many seeded sample paths; samples are illustrations, not proofs or an enumeration of compatible histories.

The diffusion panel exposes $x_0, b, \kappa, T, t/T, R$ and switches between exact and Gaussian noisy endpoint observations. The finite-depth doubling panel illustrates branch labels while stating the infinite-history limitation. A realizability panel reproduces the finite-support example. The topology panel displays the sphere-height fibre and a constant historical projection. Each panel states its carrier and mathematical scope.

Compile from the package root using XeLaTeX twice on each wrapper. Nimbus Roman, Sans and Mono PS fonts provide Latin and Cyrillic text; mathematical equations are shared. Running the verification script tests boundary limits, positive interior variances, the numerical RTS example, and additional positive-variance random-walk models against independent batch Gaussian conditioning. It does not replace proof review.

The English and Russian files are parallel manifestations of this version, not successive revisions. The package includes a file checksum manifest and an audit recording exact source locators, claim boundaries and render checks. Upstream DOI records identify WR-I–III; no unpublished WR-IV DOI is fabricated. There are no experimental datasets, fitted model parameters or new smoothing algorithm in this release.

References

- [1] A. A. Malachevsky. *Response-Defined World Spaces: Response Equivalence, Finite-Resolution Separation, and Identifiability of Global Realizations*. WR-I, WREH, v0.3 (2026). doi:10.5281/zenodo.23210258.
- [2] A. A. Malachevsky. *Finite Consistency and Global World Realizability*. WR-II, WREH, v0.5 (2026). doi:10.5281/zenodo.23224154.
- [3] A. A. Malachevsky. *Geometry of Admissible World Fibres: Refinement, Rigidity and Realizability Walls*. WR-III, WREH, v0.6 (2026). doi:10.5281/zenodo.23228682.
- [4] R. E. Kalman. Mathematical Description of Linear Dynamical Systems. *Journal of the Society for Industrial and Applied Mathematics, Series A: Control* 1(2) (1963), 152–192. doi:10.1137/0301010.
- [5] A. M. Stuart. Inverse problems: A Bayesian perspective. *Acta Numerica* 19 (2010), 451–559. doi:10.1017/S0962492910000061.
- [6] A. Clark and R. Fokink. Embedding solenoids. *Fundamenta Mathematicae* 181(2) (2004), 111–124. doi:10.4064/fm181-2-2.
- [7] J. W. Pitman. *Brownian Bridge*. Stat205B: Probability Theory, Lecture 22, Spring 2003, University of California, Berkeley; scribe T. Chen. https://www.stat.berkeley.edu/~pitman/s205s03/b_bridge.pdf.
- [8] H. E. Rauch, F. Tung, and C. T. Striebel. Maximum Likelihood Estimates of Linear Dynamic Systems. *AIAA Journal* 3(8) (1965), 1445–1450. doi:10.2514/3.3166.
- [9] S. Särkkä and L. Svensson. *Bayesian Filtering and Smoothing*, 2nd ed. Cambridge University Press, 2023. doi:10.1017/9781108917407.
- [10] P. Selick. *Fibre Bundles; Characteristic Classes; K-theory*. Undated lecture notes, University of Toronto, Section 1.2, p. 7. <https://www.math.toronto.edu/selick/fibre.pdf>.
- [11] M. Simons and J. Washburn. *Finite-Resolution Identifiability and Measurement Design for Molecular Conformer Spectroscopy*. arXiv:2608.02637v1 (2026). doi:10.48550/arXiv.2608.02637.